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Galaxy Classification for Next-Generation Astronomical Surveys Proposal

Xiaofeng Liang

M.DataSc.

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List of abbreviations

AWS	Amazon Web Services
CNN	Convolutional Neural Network
DESI	Dark Energy Spectroscopic Instrument
EC2	Elastic Compute Cloud
EDA	Exploratory Data Analysis
FITS	Flexible Image Transport System
FN	False Negatives
FP	False Positives
GPU	Graphics Processing Unit
HPC	High Performance Computing
JPEG	Joint Photographic Experts Group
LSST	Legacy Survey of Space and Time
MJD	Modified Julian Date
PCA	Principal Components Analysis
PDF	Portable Document Format
PII	Personal Identifiable Information
SAS	Science Archive Server
SDSS	Sloan Digital Sky Survey
SQL	Structured Query Language
TB	Terabyte
TN	True Negatives
TP	True Positives
URL	Uniform Resource Locator

Chapter 1: Introduction

This project proposes that the challenge of galaxy classification in an era of unprecedented astronomical data can be effectively addressed by a scalable, multi-modal machine learning framework. The exponential growth of data from modern surveys has rendered traditional, manual methods impractical, creating an urgent need for automated and reliable classification to advance our understanding of the universe. This proposal outlines a solution that leverages the full spectrum of available data—including images, spectra, and tabulated photometric information—to build a more accurate and robust system. This document will first establish the context by reviewing the current methods for galaxy classification, identifying the specific problems with these techniques. Following this, it will detail the project's objectives, the proposed multi-modal solution, and the rigorous evaluation strategy.

Chapter 2: Current Situations

This section reviews the current situation and methods that have been used in galaxy classification. Galaxies are classified based on their shapes and structures, which are essential for the understanding of the formation and evolution of galaxies in the universe [1]. This research area has a long history that began with Hubble's visual classification of galaxies in 1926, where 400 galaxies were classified by hand into categories such as elliptical, spiral, and irregular [2]. Since then, numerous refinements and extensions of this framework have been proposed, but the two most traditional approaches remain visual inspection and Principal Components Analysis (PCA). Visual inspection involves experts examining observational data, such as galaxy images and spectra, to identify recognisable patterns by eye and through their knowledge. Astronomers may assess morphological characteristics, including spiral arms, bulges, or ellipticity, as well as spectral features that provide insights into stellar populations [3]. This approach captures subtle details but relies heavily on the experience of the observers. In contrast, PCA is a statistical technique that reduces the dimensionality of large astronomical datasets by transforming them into a set of orthogonal components, each capturing a portion of the variance [4]. In galaxy classification, PCA has been widely applied to quantify photometric and spectroscopic properties such as brightness, colour indices, and spectral line strengths [5]. By projecting galaxies onto principal components, astronomers can separate populations according to star formation activity, stellar age, or overall morphology. This enables a more systematic treatment of massive surveys, where visual inspection alone would be impractical.

The paradigm shifted with the arrival of large-scale digital surveys, such as the Sloan Digital Sky Survey (SDSS), which made purely manual classification impossible [6]. This challenge led to the creation of the Galaxy Zoo project, a citizen science initiative that produces vast, human-labelled datasets essential for training supervised machine learning algorithms. More recently, the field has been revolutionised by deep learning, with Convolutional Neural Networks (CNNs) proving highly effective at classifying galaxies from image data. The current frontier is multi-modal learning, which seeks to integrate images with spectroscopic and photometric data for a more holistic analysis. These modern, data-driven approaches now represent the state-of-the-art in the field.

Chapter 3: Problem Identification

This section discusses the limitations of both traditional visual inspection and PCA-based galaxy classification methods. First, visual inspection depends on human judgement, which is inherently subjective and becomes inefficient with the large data volumes from modern surveys. The growing amount of data makes visual inspection less practical; for instance, the Sloan Digital Sky Survey (SDSS) has observed 1 million galaxies since 2000 [7]. Additionally, newer instruments like the Dark Energy Spectroscopic Instrument (DESI) observed over 100,000 galaxies in a single night [8]. Other examples include the Legacy Survey of Space and Time (LSST), which generates about 15 TB of data each night [9], and the 4MOST Hemisphere Survey that will gather uniform spectroscopy and redshifts for around six million galaxies [10]. Therefore, the exponential increase in data volume undermines the practicality of visual inspection. Second, methods employing PCA face their own challenges. While PCA reduces dimensionality and highlights variance, the resulting principal components are linear combinations of original features, which makes them harder to interpret intuitively. This complicates understanding which features influence classification. Additionally, PCA assumes linear relationships, which may not suit the complex, non-linear nature of galaxy morphology [11]. Without proper data scaling, PCA can distort the importance of features and risk losing information relevant to classification [12].

Beyond these foundational issues, even state-of-the-art deep learning approaches present unresolved problems:

- The challenge of Effective Multi-Modal Fusion: While astronomical surveys provide rich, multi-modal data, the optimal method for fusing these heterogeneous data streams (images, spectra, and tabular data) is an active area of research [13]. Most current models are unimodal (using only images), failing to leverage the full context available for each galaxy.
- The problem of Domain Shift: A model trained on data from one survey (the "source domain", e.g., SDSS) often performs poorly when applied to another survey (the "target domain", e.g., LSST) due to differences in instrumentation and data processing [14]. The "domain shift" prevents the creation of truly universal classification tools.
- The interpretability challenge: Deep learning models are often criticised as "black boxes", making it difficult to understand which features drive their classifications [15]. This lack of transparency is a significant drawback for scientific applications where understanding the reasoning is as important as the prediction itself.

Chapter 4: Objectives and Deliverables

This project aims to deliver a robust machine learning framework for galaxy classification by achieving the following primary objectives:

- To curate and preprocess a comprehensive multi-modal galaxy dataset, integrating image, spectroscopic, and tabulated photometric data.
- To establish robust performance benchmarks by developing and evaluating baseline models for each individual data modality.
- To design and implement a novel hierarchical multi-modal deep learning architecture that effectively fuses the different data types using an attention-based mechanism.
- To demonstrate the superiority of the multi-modal framework by showing a significant performance increase over the single-modality baselines on a defined morphological classification task.
- To validate the framework's versatility by demonstrating its capability to perform both morphological (shape-based) and physical (e.g., star-formation-based) galaxy classification.

Chapter 5: Proposed Solution

This section proposes a comprehensive solution to the scalability and consistency challenges in galaxy classification by developing a hierarchical multi-modal deep learning architecture. This project will use a rich dataset of approximately 460,000 galaxies from the Sloan Digital Sky Survey (SDSS) for which image, spectral, and tabulated data are all available. The classification tasks will be grounded in the scientifically validated Galaxy Zoo classification scheme [16]. This ensures the project addresses a meaningful astronomical problem by using labels derived from millions of human classifications, which have been extensively documented and used in peer-reviewed research. The core advantage of this approach is its ability to synthesise complementary information: images reveal visual structure, spectra uncover physical processes like star formation, and tabulated data provides statistical context. By combining these modalities with an attention mechanism, the model can learn to weigh the evidence from each source, leading to more accurate and robust classifications than any single-modality approach.

Chapter 6: Methods for Developing the Program

The development of the machine learning framework will follow a structured, multi-stage process, from data acquisition and definition of classification tasks to a phased model development and evaluation strategy.

6.1 Data Sources

The methodology begins with the acquisition and preparation of a multi-modal dataset sourced entirely from SDSS and its associated citizen science project, Galaxy Zoo. The dataset is constructed by programmatically downloading and joining three distinct data types for each galaxy:

Tabular Data: The foundational catalogue is created by executing a comprehensive SQL query against the SDSS Data Release 19 (DR19) CasJobs server. This query produces a high-quality cross-matched sample by joining photometric, spectroscopic, and Galaxy Zoo voting data, applying several quality filters to ensure the reliability of the sample. The detailed implementation of the SQL query is provided in Appendix A.1. Table 1 describes the key columns retrieved by this query.

Image Data: For each galaxy in the tabular catalogue, a corresponding JPEG image cutout is downloaded from the DESI Legacy Imaging Surveys viewer. These images provide three-colour composite views using g, r, and z bands, with a pixel scale of 0.1 arcseconds per pixel. The URL construction methodology using the celestial coordinates is detailed in Appendix A.2.

Spectral Data: The 1D spectral data for each galaxy is obtained as FITS files from the SDSS DR19 Science Archive Server (SAS). These spectra contain wavelength-calibrated flux measurements that reveal physical properties such as star formation rates and stellar populations. The methodology for constructing download URLs using plate, MJD, and fibre identifiers is provided in Appendix A.3.

Figure 1 shows an example of DR19 data records illustrating how the three data modalities (image, spectrum, and tabular data) are linked for a single galaxy.

Table 1: Description of Key Tabular Data Columns

Category	**Column Name**	**Description**
Identifiers	`objid`	The primary unique identifier for each object in the SDSS catalogue.
	`specObjID`	The unique identifier for a specific spectroscopic observation.
	`plate`, `mjd`, `fiberid`	A combination of three values that uniquely identifies an SDSS spectrum. Used to construct the download URL for the spectral FITS file.
Positional Data	`ra`, `dec`	The celestial coordinates (Right Ascension, Declination) in degrees. Used to construct the download URL for the corresponding image cutout.
Vote Counts	`nvote_tot`	The total number of votes cast by Galaxy Zoo volunteers for a given galaxy, indicating the reliability of the classification.
Vote Fractions	`p_el`	The fraction of votes classifying the galaxy as Elliptical.
	`p_cw`	The fraction of votes classifying the galaxy as a Clockwise Spiral.
	`p_acw`	The fraction of votes classifying the galaxy as an Anticlockwise Spiral.
	`p_edge`	The fraction of votes classifying the galaxy as an Edge-on Disk.
	`p_dk`	The fraction of votes classifying the galaxy as Don't Know / Artifact.
	`p_mg`	The fraction of votes classifying the galaxy as a Merger.
	`p_cs`	A convenience column representing the combined fraction of votes for any spiral type (`p_cw` + `p_acw` + `p_edge`).

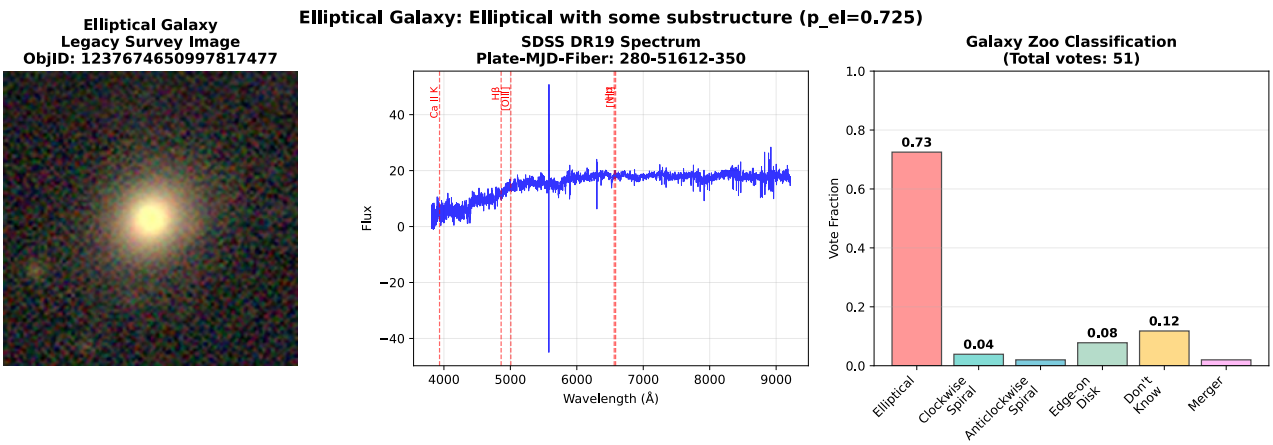

Figure 1: Visualisation of Sample Data from SDSS DR19

6.2 Classification Tasks

From this integrated, multi-modal dataset mentioned in the last section, a hierarchy of classification tasks is defined as follows:

- **Primary Task: 4-Class Simplified Morphology:** This will be the main focus for development and benchmarking. Labels are generated from the Galaxy Zoo vote fractions by assigning each galaxy to the class (Elliptical, Spiral, Merger, or Uncertain) with the highest volunteer vote count, using the `p_el`, `p_cs`, `p_mg`, and `p_dk` columns, respectively.
- **Advanced Task 1: 6-Class Full Morphology:** As a stretch goal, the model will be trained on the more granular 6-class scheme, separating clockwise (`p_cw`) and anticlockwise (`p_acw`) spirals from edge-on disks (`p_edge`).
- **Advanced Task 2: Physical Classification:** To demonstrate the full power of the multi-modal approach, a 2-class physical classification task (Star-Forming vs. Passive) will be implemented by deriving labels from the downloaded spectroscopic data.

6.3 Model Development: A Phased Approach

Model development will proceed progressively to systematically build complexity and validate the contribution of each modality. This approach is designed to first understand the predictive power of each data type in isolation before combining them into a more powerful, integrated system. The model will be developed in three phases.

6.3.1 Phase 1: Exploratory Data Analysis

This phase focuses on understanding the data and establishing robust performance benchmarks. The insights gained from the EDA will directly inform the modelling strategy. For example, the EDA will quantify the class imbalance in the dataset, which will necessitate the use of evaluation metrics like the Macro-Averaged F1-Score and may require techniques like weighted loss functions during training.

6.3.2 Phase 2: Single-Modality Baselines

Three separate baseline models will be developed, with each model chosen for its suitability to a specific data type. The first baseline model is **tabular-only**. For the structured, tabular data containing photometric properties and Galaxy Zoo vote fractions, tree-based ensemble models like Random Forest [17] and XGBoost [18] are the industry standard. They are highly effective

at capturing complex, non-linear relationships between features and are robust to outliers. They also provide feature importance scores, which will be valuable for understanding the predictive power of different tabular features [19].

The second baseline model is **image-only**. For the image data, a pre-trained Convolutional Neural Network (CNN), such as a ResNet architecture, will be used. CNNs are the state-of-the-art for image classification tasks as their convolutional layers are specifically designed to learn hierarchical spatial features like edges, textures, and complex shapes (e.g., spiral arms) [20]. Using a pre-trained model leverages transfer learning, which significantly accelerates training and improves performance by starting with a network that has already learned to recognise basic visual features from a massive dataset like ImageNet.

The third model is **spectral-only**. The 1D spectral data, which represents light intensity across a range of wavelengths, is effectively a sequence. A 1D-CNN is the ideal choice for this modality because its 1D convolutional filters are designed to slide along the sequence and learn to recognise characteristic patterns, such as the shape and location of specific emission or absorption lines that are critical for physical classification [21].

6.3.3 Phase 3: Triple Fusion - The Full Multi-Modal Architecture

The innovative core of this project is the development of a hierarchical multi-modal architecture designed to intelligently integrate all three data streams. This architecture consists of two key stages: modality-specific feature extraction and attention-based fusion. First, each data type will be processed by its own specialised encoder, as described in Phase 2, to extract a high-level feature vector that summarises the most important information from that modality. Second, instead of simply concatenating these feature vectors, they will be fused using a **cross-modal attention mechanism**. This is a critical design choice because an attention mechanism allows the model to dynamically learn the relative importance of each data modality for a given classification [22]. For example, for a clear, face-on spiral galaxy, the model might learn to "pay more attention" to the image features, while for a galaxy with ambiguous visual features but a strong star-forming signature in its spectrum, it might learn to weigh the spectral features more heavily. This approach not only improves classification accuracy but also provides a degree of interpretability, as the resulting attention maps can be used to understand the model's reasoning, which has been shown to align with the regions an expert human classifier would focus on.

6.4 Model Evaluation Strategy

A rigorous and quantitative evaluation strategy is essential for demonstrating the project's success. The performance of all models will be assessed on a held-out test set that is not used during training or hyperparameter tuning.

For a given class, well-known definitions are utilised, including True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). The primary evaluation metrics are:

- **Accuracy:** The overall fraction of correct predictions

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{6.1}$$

- **Precision:** Of all the galaxies the model predicted for a certain class, what fraction was correct? High precision means a low false positive rate.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6.2}$$

- **Recall (Sensitivity):** Of all the galaxies that truly belong to a certain class, what fraction did the model correctly identify? High recall means a low false negative rate.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{6.3}$$

- **F1-Score:** The harmonic mean of Precision and Recall, providing a single score that balances the two.

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{6.4}$$

For the multi-class problem, these metrics will be calculated on a per-class basis. The primary performance indicator for this project will be the **Macro-Averaged F1-Score**. This metric is calculated by taking the unweighted average of the F1-Scores for each class. This is the most appropriate metric for this project because it treats each class equally, regardless of its size in the dataset [23]. This ensures that the model's performance on rare but scientifically important classes is given the same importance as its performance on more common classes, providing a true measure of the model's overall classification capability.

Chapter 7: Resources

Successful completion of this project requires specific software, hardware, and data resources. The project will be developed in Python 3.12, utilising standard data science libraries (`Pandas`, `NumPy`, `Scikit-learn`), deep learning framework (`PyTorch`), and astronomical packages (`astropy`). While initial development can be done on a standard laptop, the computationally intensive model training will require access to a cloud platform providing High Performance Computing (HPC), like Amazon SageMaker. All data will be sourced from the publicly available SDSS, incurring no direct cost.

Chapter 8: Schedule

The project is designed to be completed within a 12-week semester. The following Gantt chart (Figure 2) outlines the schedule, incorporating the phased model development strategy.

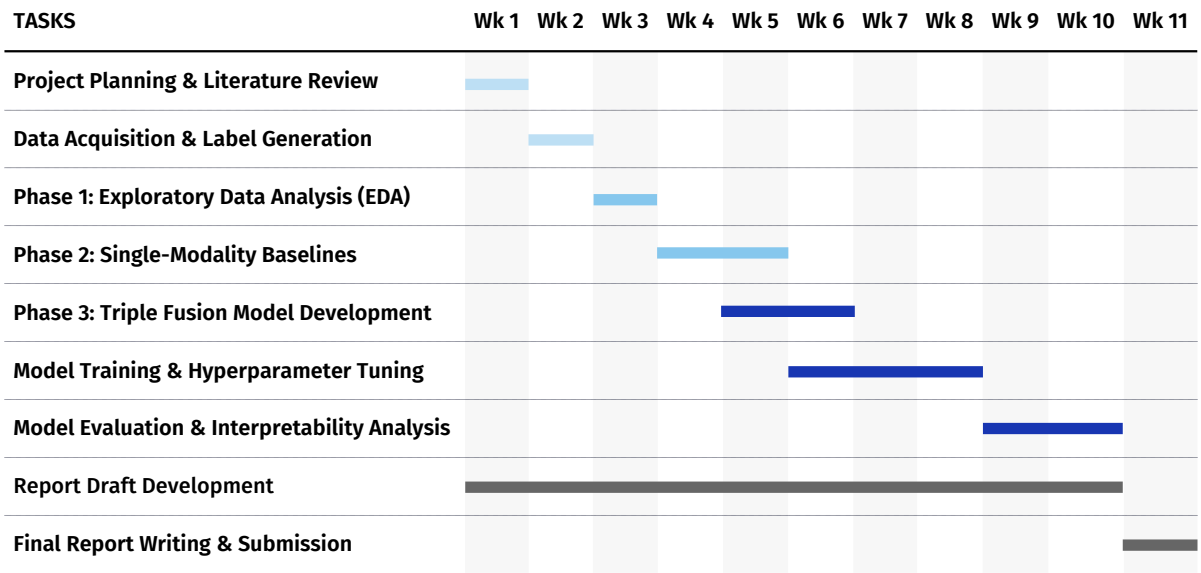

Figure 2: Proposed Project Schedule

Chapter 9: Costs

The primary cost of this project remains the time commitment of the researcher, as detailed in the project schedule. However, the computationally intensive task of training deep learning models will use cloud-based GPU instances from Amazon Web Services (AWS), which introduces a potential financial cost. Two primary approaches were considered: direct usage of Amazon EC2 instances and the fully managed Amazon SageMaker platform [24]. Table 2 provides a detailed comparison of these two approaches.

Table 2: Comparison of AWS Training Approaches

Feature	**Amazon EC2 (Direct Usage)**	**Amazon SageMaker**
Control and Flexibility	Maximum control over the operating system, drivers, and specific software versions. Ideal for highly customised environments.	Managed environment that simplifies standard workflows. High control is possible via custom containers, but requires more setup.
Management Overhead	High. The user is responsible for manual instance setup, software installation, security patching, and crucially, shutting down instances to avoid costs.	Low. SageMaker automates instance provisioning, job execution, and termination, significantly reducing management overhead and the risk of incurring unnecessary costs.
Cost Model	Lower base hourly rate for raw compute instances.	Carries a premium over the base EC2 cost for its management features, but provides powerful, integrated cost-saving tools.

For this project, the recommended approach is Amazon SageMaker. While the base instance cost is slightly higher, the reduction in management overhead is a significant advantage for a time-constrained academic project. More importantly, SageMaker's built-in features for cost optimisation are more accessible and directly suited to machine learning workflows.

The following cost-mitigation strategies will be employed:

- **Leverage Managed Spot Training:** The project will primarily use SageMaker's Managed Spot Training feature, which can reduce training costs by up to 90% compared to on-demand instances by using spare EC2 capacity. SageMaker's implementation is particularly beneficial as it automatically manages interruptions and can resume training from checkpoints.
- **Utilise Free Tier:** The project will leverage AWS Free Tier to obtain service credits, which can offset any remaining costs from the training jobs.

- **Right-size Compute Instances:** The project will start with smaller, cost-effective GPU instances (e.g., from the `ml.g4dn` family) and only scale up if required, ensuring that resources are not over-provisioned.

By implementing these strategies, the direct financial cost of the required computational resources is expected to be minimal.

Chapter 10: Qualifications

As a Master of Data Science student, I have a comprehensive skill set that directly aligns with the technical demands of this project. My qualifications ensure that each phase of the proposed work can be executed efficiently and to a high standard:

- **Proficiency in Python and Data Science Libraries:** I possess advanced skills in Python and its core data science ecosystem, including Pandas for data manipulation, `NumPy` for numerical operations, and `Matplotlib`/`Seaborn` for data visualisation. This expertise is essential for scripting the entire data acquisition pipeline, performing the Exploratory Data Analysis (EDA), and preparing the data for modelling.
- **Machine Learning and Deep Learning Expertise:** Through graduate-level coursework, I have gained in-depth knowledge of traditional machine learning algorithms and a basic level of deep learning architectures. My experience with Scikit-learn is directly applicable to building the baseline models (Random Forest, XGBoost). At the same time, my familiarity with `PyTorch` is crucial for designing and implementing the novel multi-modal CNN architecture.
- **Cloud Computing Competency:** I am an AWS Certified Cloud Practitioner and AWS Certified Associate Solutions Architect, which provides me with the foundational knowledge to manage cloud resources effectively. This certification demonstrates my ability to provision, monitor, and optimise cloud services, ensuring that the model training phase on AWS SageMaker will be conducted cost-effectively and efficiently. This qualification directly supports the cost-mitigation strategies outlined in the previous section.

These qualifications provide me with the necessary technical foundation to successfully navigate the challenges of this project, from data engineering and EDA to advanced deep learning.

Chapter 11: Risk Management, Ethics, and AI Disclosure

This project will be conducted with a proactive approach to management, a clear understanding of ethical obligations, and transparent disclosure of AI tool usage.

11.1 Risk Management

A comprehensive plan is in place to identify and mitigate potential risks to the project's timeline and success. There are two main risks. The first one is data quality issues, such as corrupted image files, noisy spectra, or significant class imbalance in the training data. To mitigate this risk, robust data validation scripts will be implemented to check for and exclude corrupted files. If class imbalance proves to be a significant issue, techniques such as data augmentation and the use of weighted loss functions will be employed to ensure the model does not become biased towards dominant classes. The second risk is that the proposed multi-modal deep learning model fails to converge during training or does not outperform the single-modality baselines. To handle this potential risk, establish strong baseline models. If the full fusion model underperforms, the pairwise fusion models from Phase 2 will provide insights into which modality combinations are most effective, allowing for a targeted redesign of the architecture.

11.2 Ethical Consideration

The ethical considerations for this project are straightforward. The project exclusively utilises publicly available, anonymised scientific data from large-scale astronomical surveys (SDSS, DESI) and the Galaxy Zoo citizen science project. The data pertains to celestial objects and does not involve human subjects, personal identifiable information (PII), or any form of sensitive or proprietary data. Therefore, there are no ethical concerns regarding privacy, data ownership, or consent. The project operates well within the established norms of open scientific research.

11.3 AI Disclosure

In accordance with university policy, the use of generative AI tools in the preparation of this proposal is transparently disclosed. Generative AI tools were utilised to assist with sentence paraphrasing, grammar correction, and brainstorming to improve the clarity and readability of this document. The core scientific concepts, the methodological design, the data acquisition plan, and the final analysis remain the original work of the author.

Chapter 12: Conclusion

This proposal outlines a plan to build and evaluate a machine learning model that classifies galaxies by combining three different types of data: images from the DESI Legacy Survey, spectra from the SDSS, and tabulated classifications from the Galaxy Zoo project.

Astronomical surveys now produce far too much data for astronomers to classify by hand, and current automated methods often fail to use all the available information for each galaxy. This project directly addresses that problem. Using a dataset of approximately 460,000 galaxies, the first step will be to build separate models for each data type to establish clear performance benchmarks.

The main goal is to then build a single, multi-modal deep learning model that fuses these three data sources. Using an attention mechanism, the project will also investigate *how* the model makes its decisions, identifying which data type is most important for classifying different kinds of galaxies.

The outcome will be a trained model and a clear, data-backed answer to whether combining multiple data types provides a better way to classify galaxies. This work will provide a practical framework for handling complexi-modal datasets that are becoming common in astronomy and other scientific fields.

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Appendix A: Technical Implementation Details

This appendix provides the technical implementation details for the data acquisition methodology described in Chapter 6.

A.1 SDSS DR19 Data Extraction Query

The following SQL query is executed against the SDSS Data Release 19 (DR19) CasJobs server to extract the foundational catalog of galaxies.

```sql
SELECT DISTINCT
    p.objid, p.ra, p.dec, s.ra, s.dec, p.l, p.b, s.specObjID, s.plate,
    s.mjd, s.fiberid, s.programname, s.class, s.sdssPrimary, v.*
INTO mydb.DATA7901_DR19
FROM PhotoObj AS p
    LEFT JOIN SpecObj AS s ON s.bestobjid = p.objid
    LEFT JOIN zooVotes AS v ON v.objid = p.objid
WHERE
    p.petroMag_r BETWEEN 10.0 AND 17.7
    AND s.zWarning = 0
    AND s.z BETWEEN 0.003333 AND 0.1500000001;
```

Listing A.1: SDSS DR19 Data Extraction Query

A.2 Image Data URL Construction

Galaxy image cutouts are obtained from the DESI Legacy Imaging Surveys viewer. For each galaxy, the image URL is constructed programmatically using the Right Ascension (`ra`) and Declination (`dec`) coordinates from the catalogue:

```
https://www.legacysurvey.org/viewer/jpeg-cutout?ra={ra}&dec={dec}
&layer=ls-dr9&pixscale=0.1&bands=grz
```

A.3 Spectral Data URL Construction

The 1D spectral data for each galaxy is downloaded as FITS files from the SDSS DR19 Science Archive Server (SAS). The URL for each spectrum is constructed using the plate (`plate`), Modified Julian Date (`mjd`), and fibre identifiers(`fiberid`):

```
https://data.sdss.org/sas/dr19/spectro/sdss/redux/26/spectra/lite/
    {plate_val}/spec-{plate}-{mjd}-{fiberid}.fits
```